

Inpatient Evaluation of Acute Dyspnea

Objectives

1. Develop a broad differential for causes of acute dyspnea



2. Construct a framework for the evaluation of acute dyspnea



3. Identify causes of acute hypoxemia

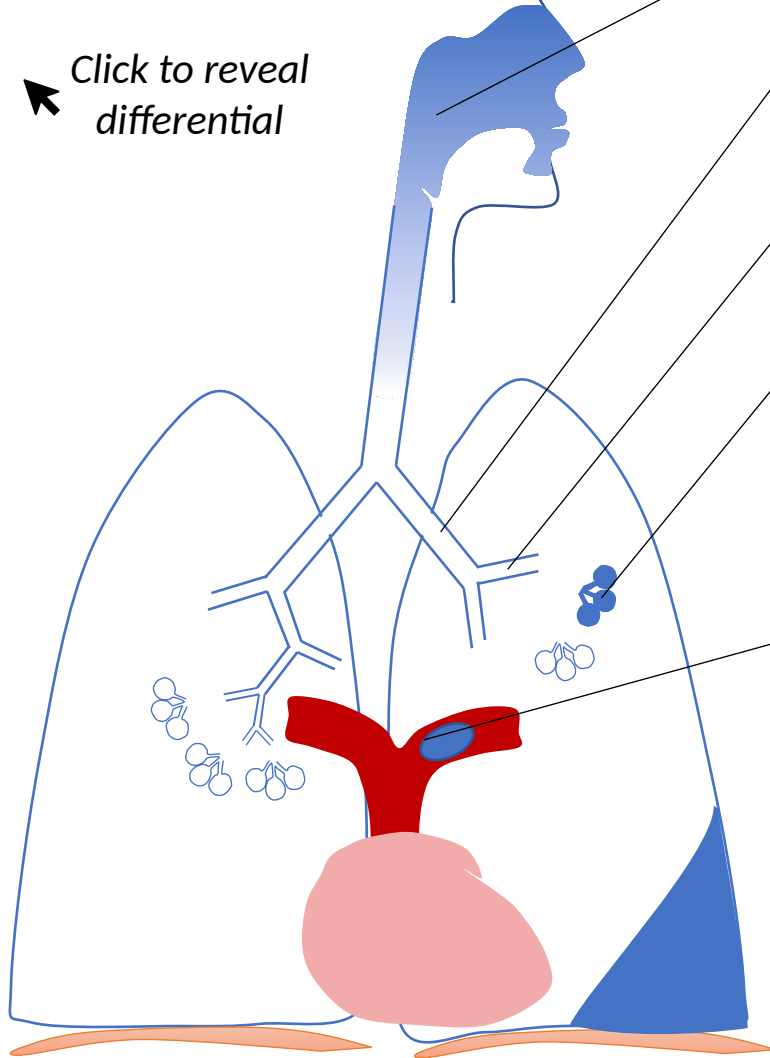


PRACTICE CASES



1. Causes of Acute Dyspnea

Click to reveal differential



Airways

Anaphylaxis, angioedema
Mucus plugging, airway stenosis
Bronchoconstriction (COPD, asthma)

Alveoli

<u>Fills alveoli</u>	<u>Collapses alveoli</u>
Blood = DAH	Atelectasis
Pus = pneumonia	Pleural process (blood, pus, water, air)
Water = edema	

Vessels

Pulmonary embolism

Diaphragm dysfunction
Abdominal pressure

Myocardial infarction
Arrhythmias Pericardial disease
Valvular disease

Anxiety
Acidosis (metabolic and respiratory)
Anemia

RESPIRATORY

Click here

Blockage or narrowing of any part of the respiratory system will cause dyspnea!

MUSCULOSKELETAL

CARDIAC

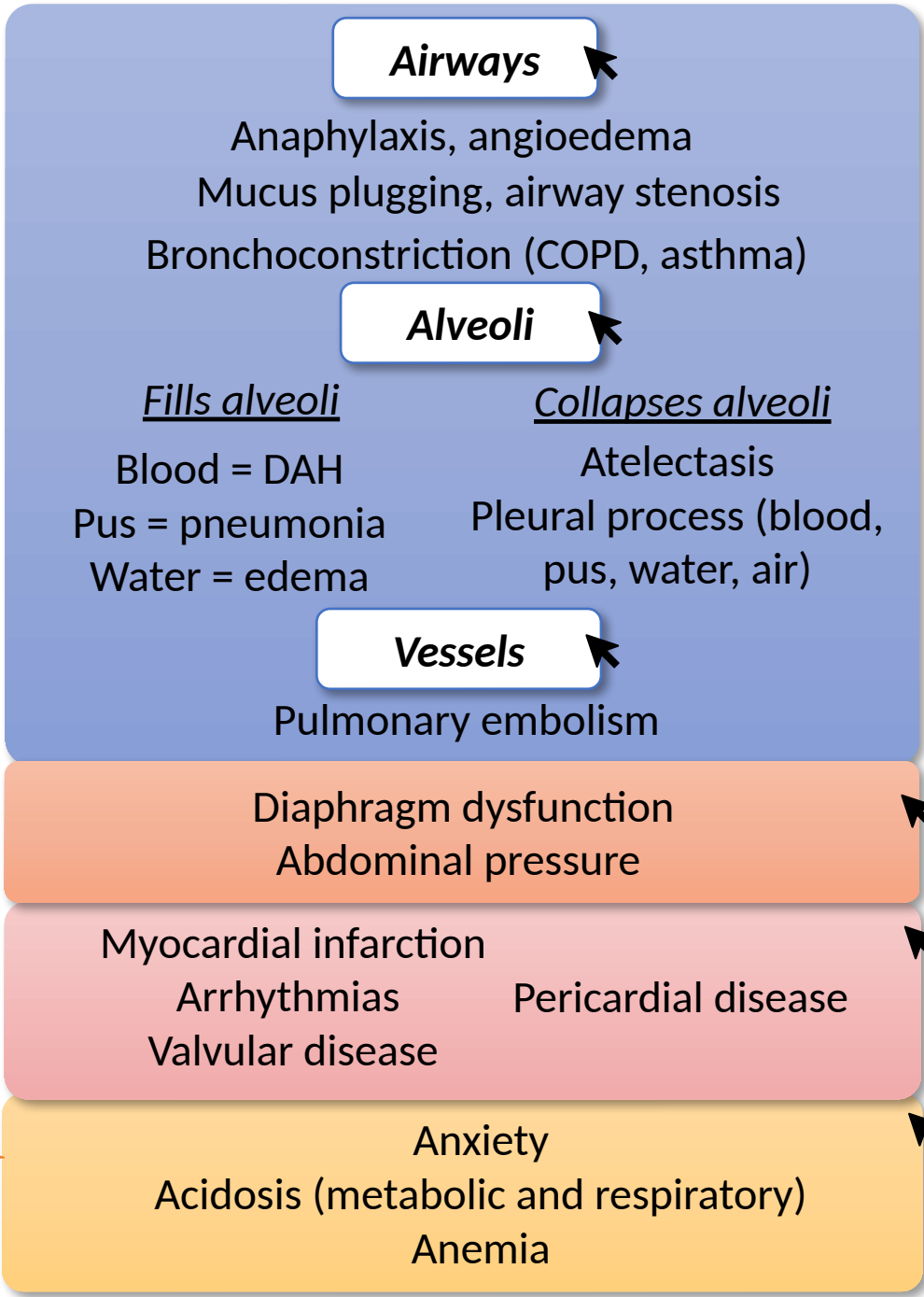
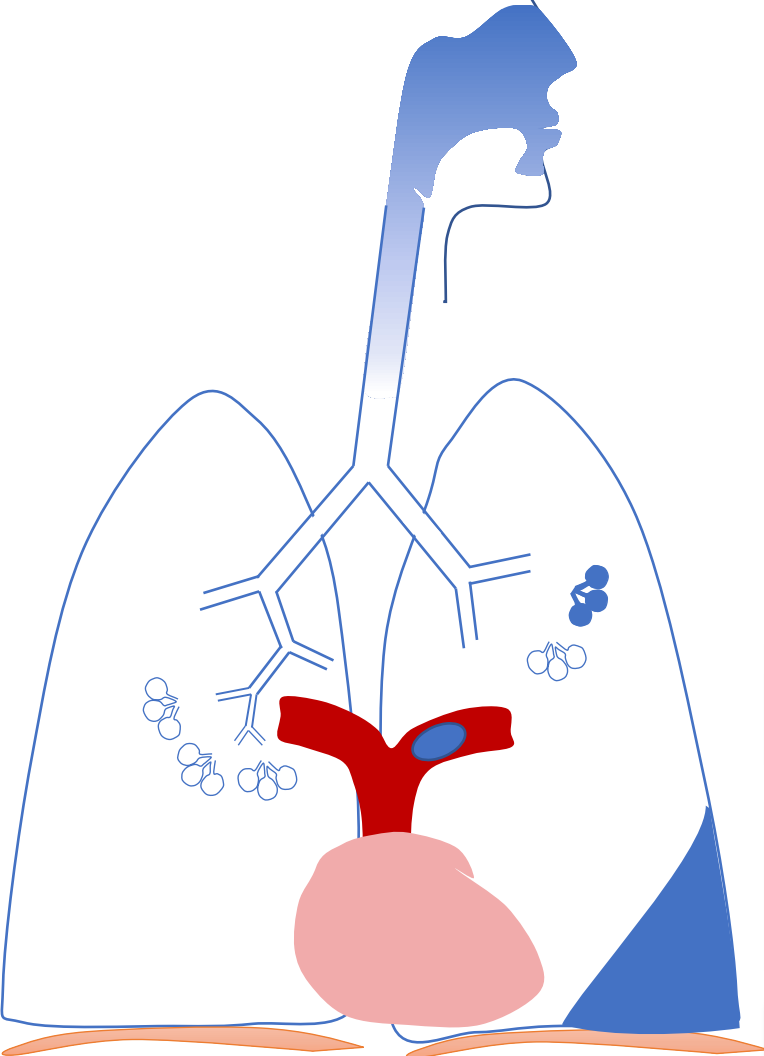
The 3 "A's"



2 EVAL

3 HYPOXEMIA

2. Evaluation and Diagnostics



How to Evaluate

Physical exam

CXR

ABG

CTPE

CXR, ABG/ VBG
Physical exam

ECG

Physical exam

BMP, ABG/ VBG
CBC

More Eval



1 CAUSES

3 HYPOXEMIA

2. Evaluation and Diagnostics

Important diagnostic tests:

CXR

ECG

CBC

ABG vs. VBG

The Age Old Question: ABG or VBG?

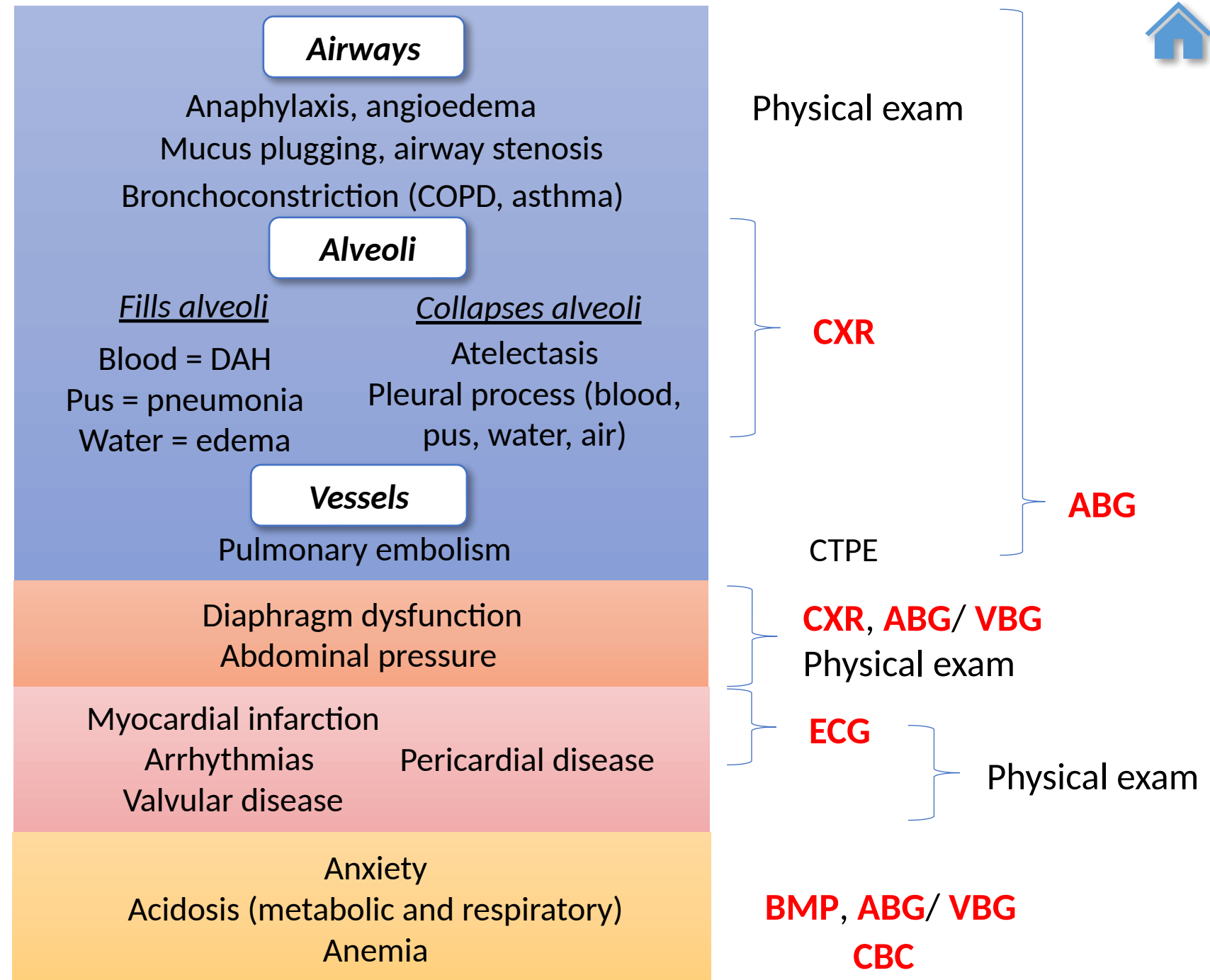
ABG: pH / $P_a\text{CO}_2$ / $P_a\text{O}_2$

VBG: pH / $P_v\text{CO}_2$

ABG pH = VBG pH + 0.04
ABG $P_a\text{CO}_2$ = VBG $P_v\text{CO}_2$ - 4

Get an ABG when you need the $P_a\text{O}_2$:

1. Calculate the A-a gradient
2. Confirming hypoxemia
3. Assessing the severity of hypoxemia



1 CAUSES

3 HYPOXEMIA

3. Hypoxemia

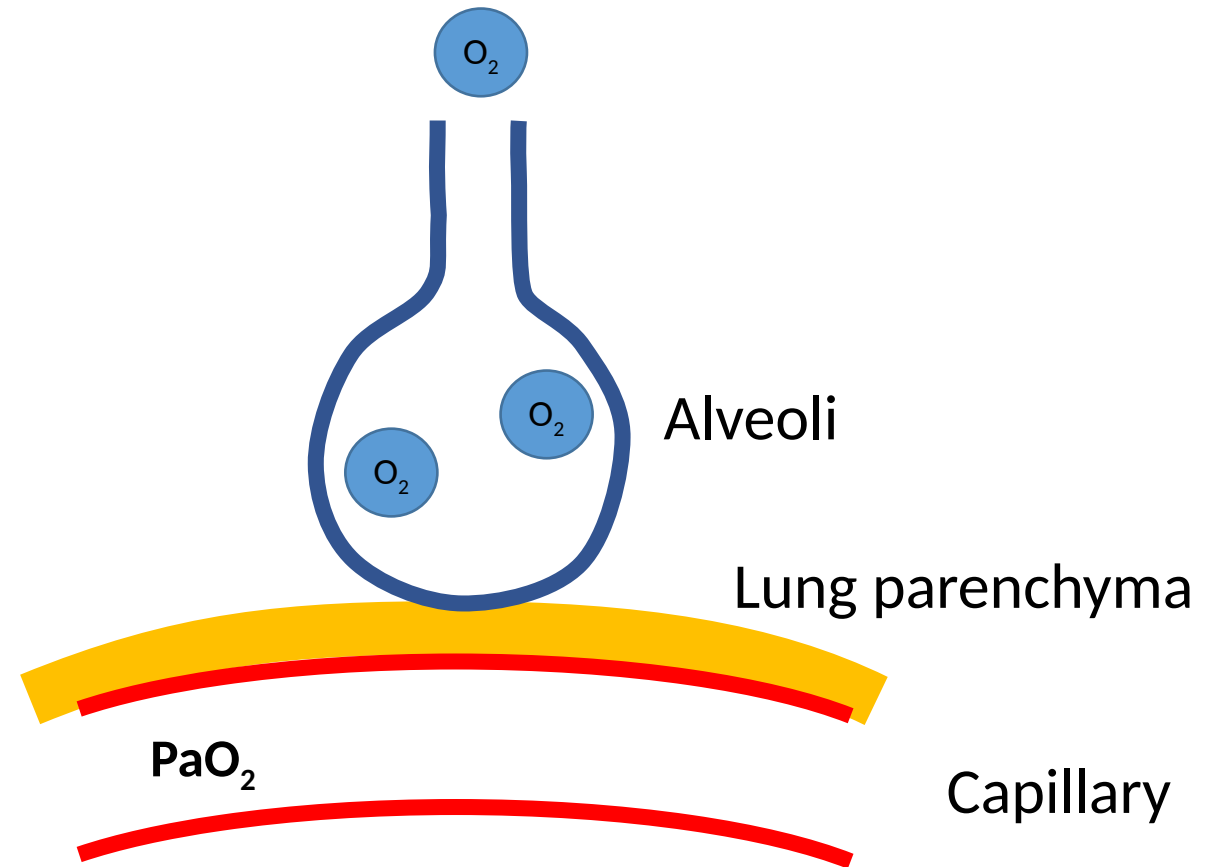
Dyspnea $\neq$ Hypoxemia

PaO_2 is measured on an ABG.

Normal PaO_2 is 75 -100.

Hypoxemia = low PaO_2

Not all causes of dyspnea cause hypoxemia.



Click Anywhere to Advance
Slide Animations and Text

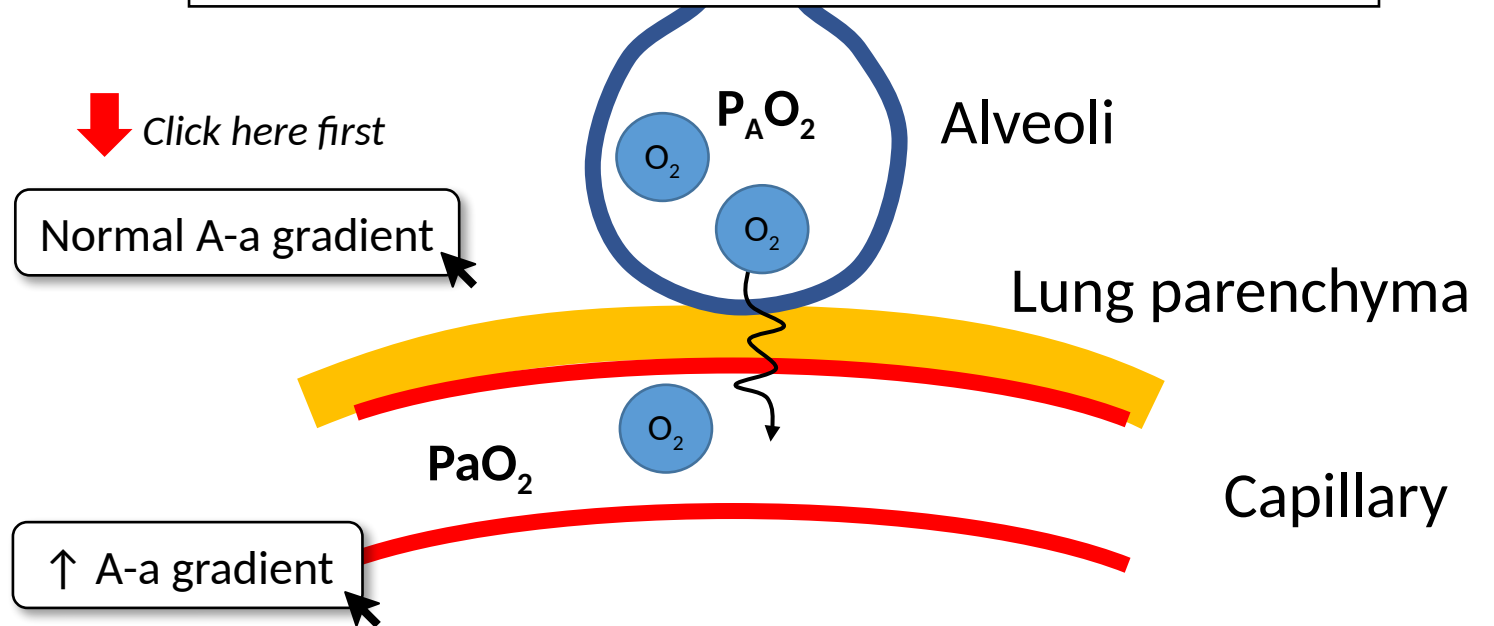
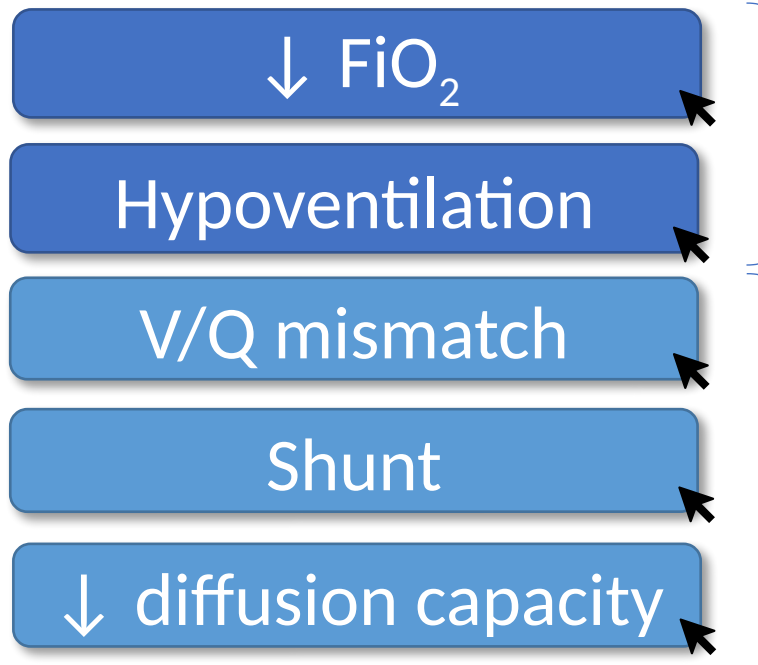


1 CAUSES

2 EVAL

Causes of Hypoxemia

There are 5 causes of hypoxemia. They can be differentiated between those that cause an $\uparrow$ A-a gradient and those that do not. Click to learn more about the A-a gradient.



An elevation in the alveolar ventilation results in a decrease in the partial pressure of CO_2 in the alveoli.	The A-a gradient = $P_A \text{O}_2 - \text{PaO}_2$ $P_A \text{O}_2 = \text{FiO}_2 \times (P_{\text{atm}} - P_{\text{H}_2\text{O}}) - P_a \text{CO}_2 (1.25)$ PaO_2 and PCO_2 are obtained from an ABG The expected A-a gradient = $\text{age} / 4 + 4$	An increase in the alveolar ventilation (hyperventilation) can lead to a decrease in the partial pressure of CO_2 in the alveoli.
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1 CAUSES

2 EVAL

3 HYPOXEMIA

Decreased FiO_2

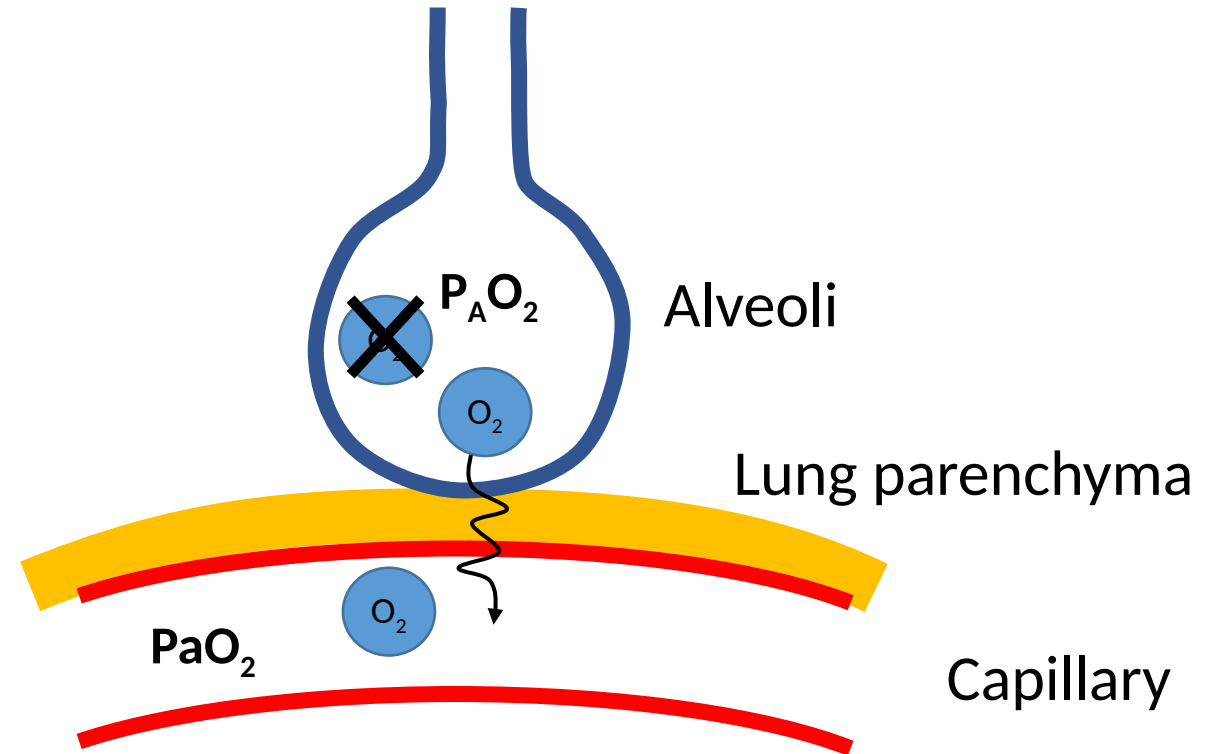
↓ FiO_2

Hypoventilation

V/Q mismatch

Shunt

↓ diffusion capacity



Decreased FiO_2 typically occurs at higher altitudes. The lower the fractional inspired concentration of O_2 , the less O_2 is available to diffuse into capillaries resulting in lower PaO_2 .

Clinically, this does not occur acutely and is not a cause of hypoxemia in the hospital.

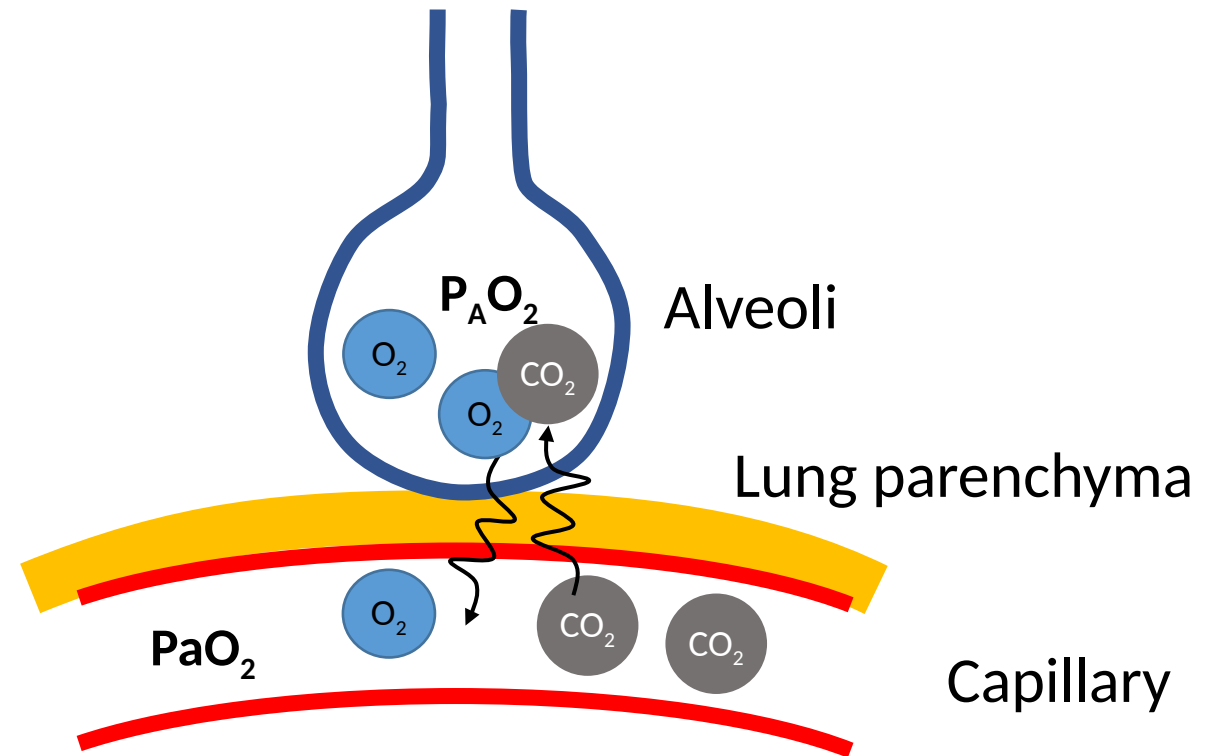
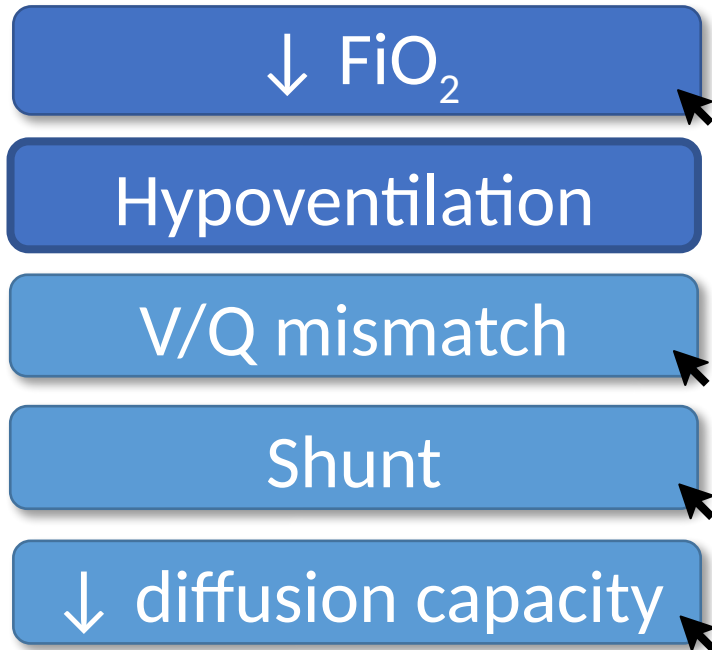


1 CAUSES

2 EVAL

3 HYPOXEMIA

Hypoventilation



Ventilation, which is determined by respiratory rate and tidal volume (volume of breaths) creates opportunities for gas exchange. Hypoventilation ↓ elimination of CO₂ resulting in ↑ PCO₂. Elevated PCO₂ effectively ↓ “space” for O₂, and subsequently results in hypoxemia. Examples include CNS depression, opiates/ benzodiazepines, neuromuscular weakness, diaphragmatic dysfunction.



1 CAUSES

2 EVAL

3 HYPOXEMIA

V/Q mismatch

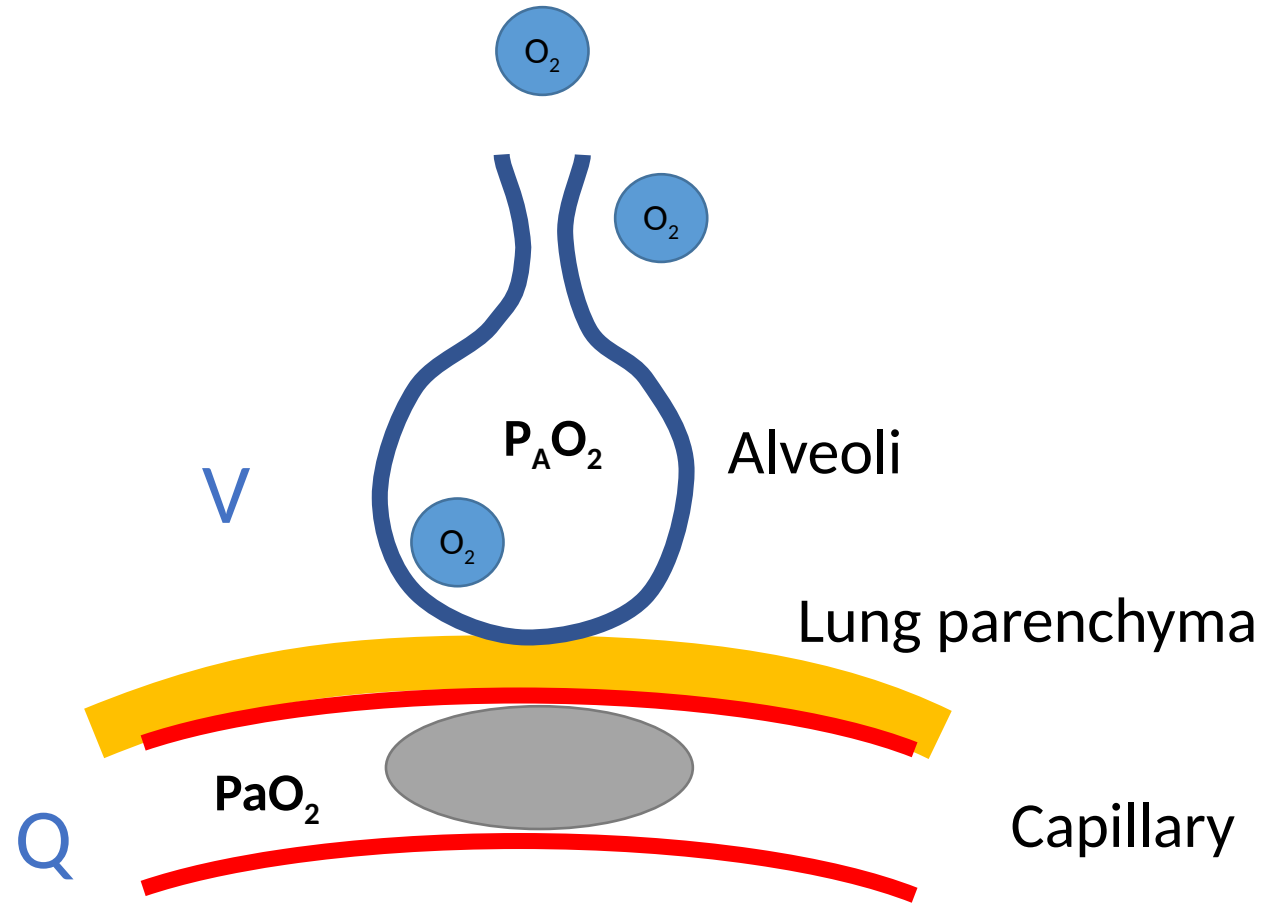
↓ FiO_2

Hypoventilation

V/Q mismatch

Shunt

↓ diffusion capacity



Ventilation (V) - Perfusion (Q) mismatch (or V/Q mismatch) occurs when there are areas of V without Q or vice versa. This can occur in the presence of smaller alveolar filling/collapsing process, obstructive airway disease, pulmonary vascular disease, or pulmonary embolism.



1 CAUSES

2 EVAL

3 HYPOXEMIA

Shunt

↓ FiO_2

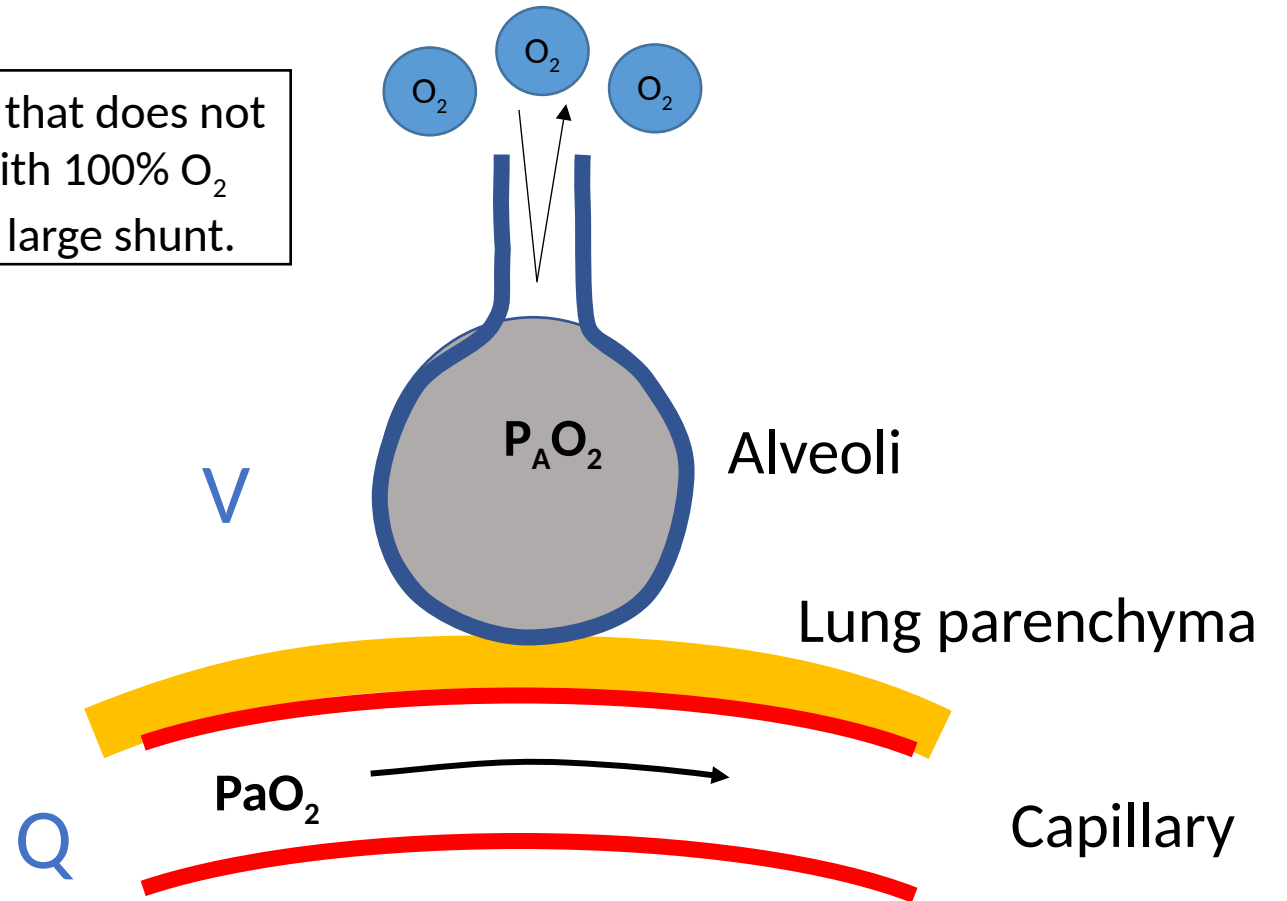
Hypoventilation

V/Q mismatch

Shunt

↓ diffusion capacity

Hypoxemia that does not correct with 100% O_2 suggests a large shunt.



Shunt occurs when there is perfusion (Q) without ventilation (V). This can occur in the presence of a physical shunt or with a large alveolar filling/collapsing process. Things that fill the alveoli include blood, water, pus.

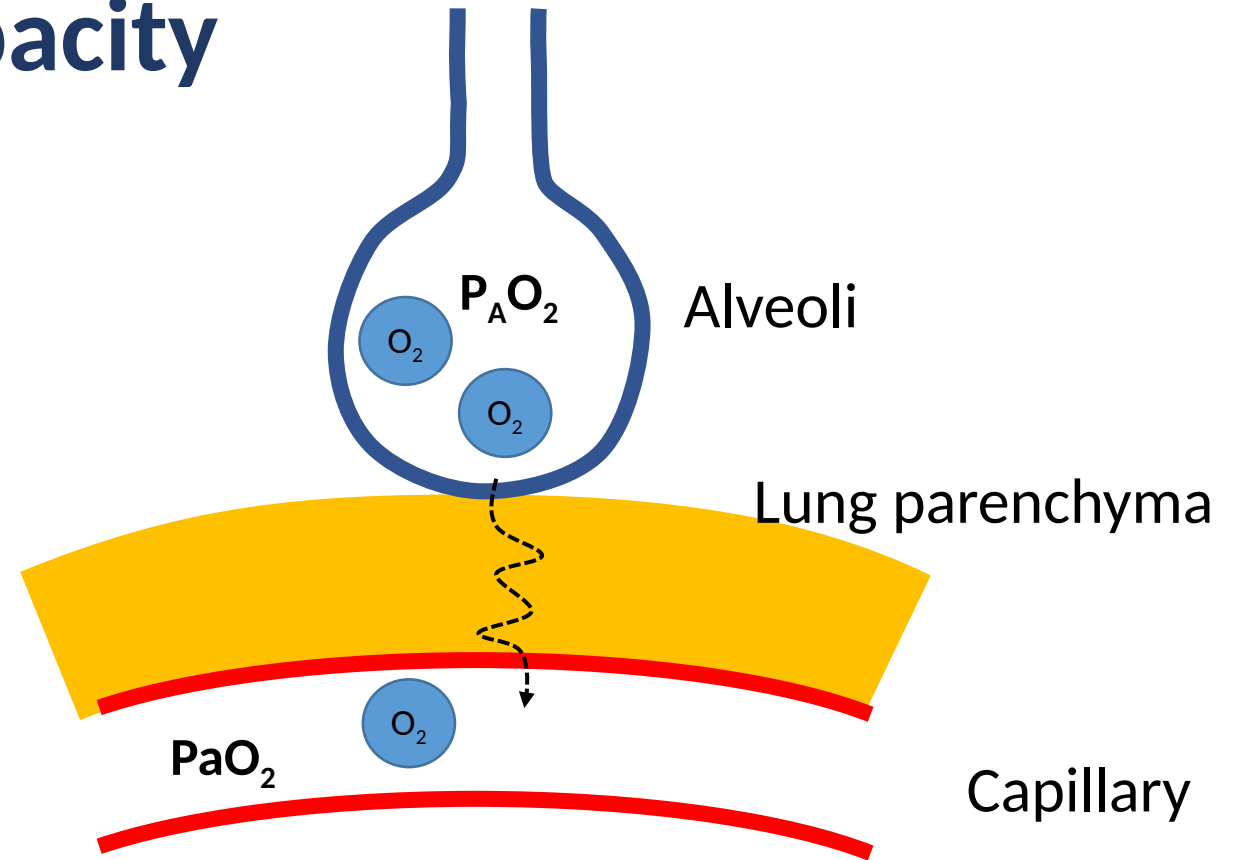
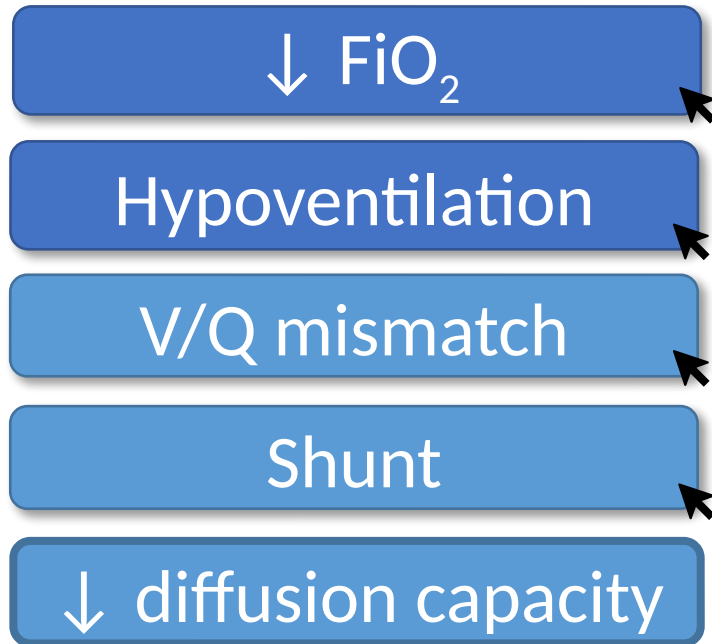


1 CAUSES

2 EVAL

3 HYPOXEMIA

Decreased diffusion capacity



Decreased diffusion capacity can occur when there is pathology in the lung parenchyma such as with interstitial lung disease/ lung fibrosis. This does not happen acutely and is unlikely a cause of acute dyspnea in the hospital.



1 CAUSES

2 EVAL

3 HYPOXEMIA

CASES

CASE 1

84 M with a history of HLD, CAD presents with 1 d of dyspnea.

- VS: T 38C, HR 110, BP 125/70, RR 24, S_aO₂ 92% on RA

• Ex

CXR looks like...



CXR

BMP/ CBC

ABG/ VBG

ECG

FINAL DIAGNOSIS

CASE 2

74 F with a history of NHL presents with 1 d of dyspnea.

- VS: T 36C, HR 110, BP 98/60, RR 24, S_aO₂ 94% on RA

• Ex

CXR looks like...



CXR

BMP/ CBC

ABG/ VBG

ECG

FINAL DIAGNOSIS

CASE 3

25 F with type 1 DM a presents with 1 d of dyspnea.

- VS: T 36C, HR 110, BP 125/70, RR 32, S_aO₂ 99% on RA

• Exam:

- Tachycardic, regular rhythm
- Tachypneic without increased work of breathing, clear lungs

Diagnostic testing

CXR

BMP/ CBC

ABG/ VBG

ECG

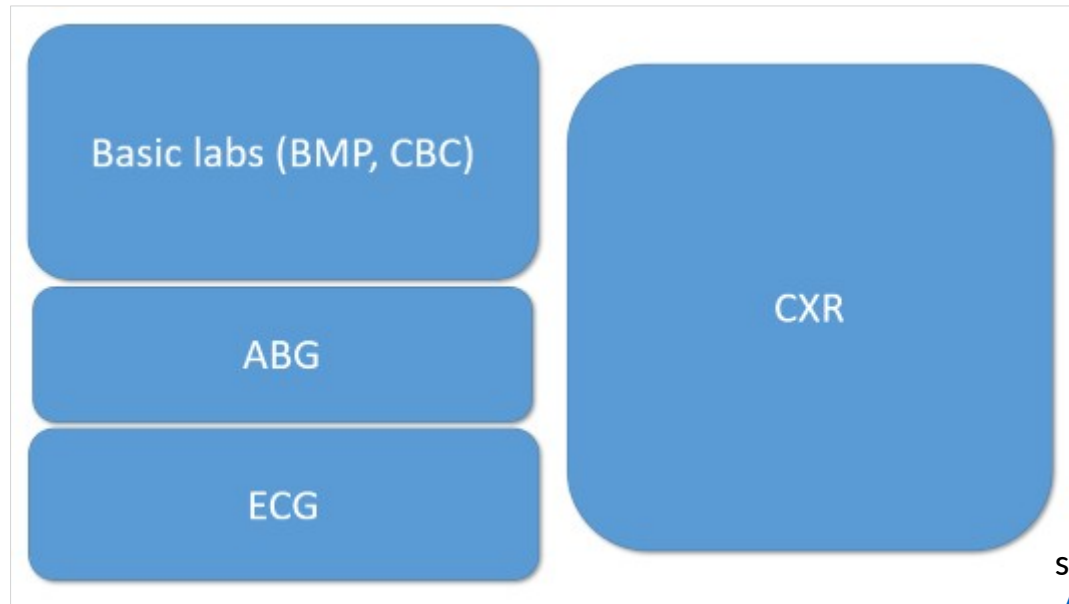
FINAL DIAGNOSIS

CASE 4 (part 1)

64 M with a history of R-sided renal cell cancer admitted for a R nephrectomy.

On post-op day 1, he developed acute onset shortness of breath and chest and abdominal pain.

- VS: T 37.1C, HR 140, BP 90/60, RR 28, S_aO_2 96% on RA
- Tachypneic, diminished lung sounds at bases. Abdomen distended and tender



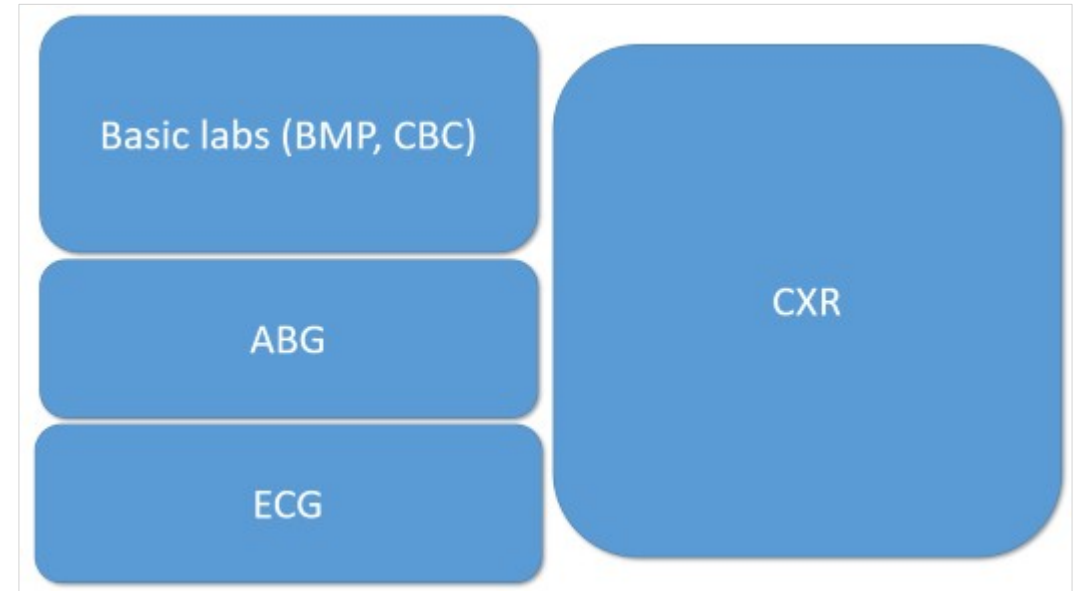
FINAL DIAGNOSIS

shows evi
[A-a gradi](#)

CASE 4 (part 2)

On post-op day 5, he developed acute onset shortness of breath and chest pain.

- VS: T 37.1C, HR 140, BP 90/60, RR 28, S_aO_2 82% on RA
- Tachypneic, lungs otherwise clear. Tachycardic without murmurs.



FINAL DIAGNOSIS



Basic labs (BMP, CBC)

ABG

PO₂, Z₁

ECG

CXR

Basic labs (BMP, CBC)

ABG

ECG

CXR

Calculate the A-a gradient

pH 7.43 / PCO₂ 32 / PO₂ 50 / HCO₃ 24 (on room air)

Calculate an A-a gradient!

$$P_AO_2 = 150 - 32 \times (1.25)$$

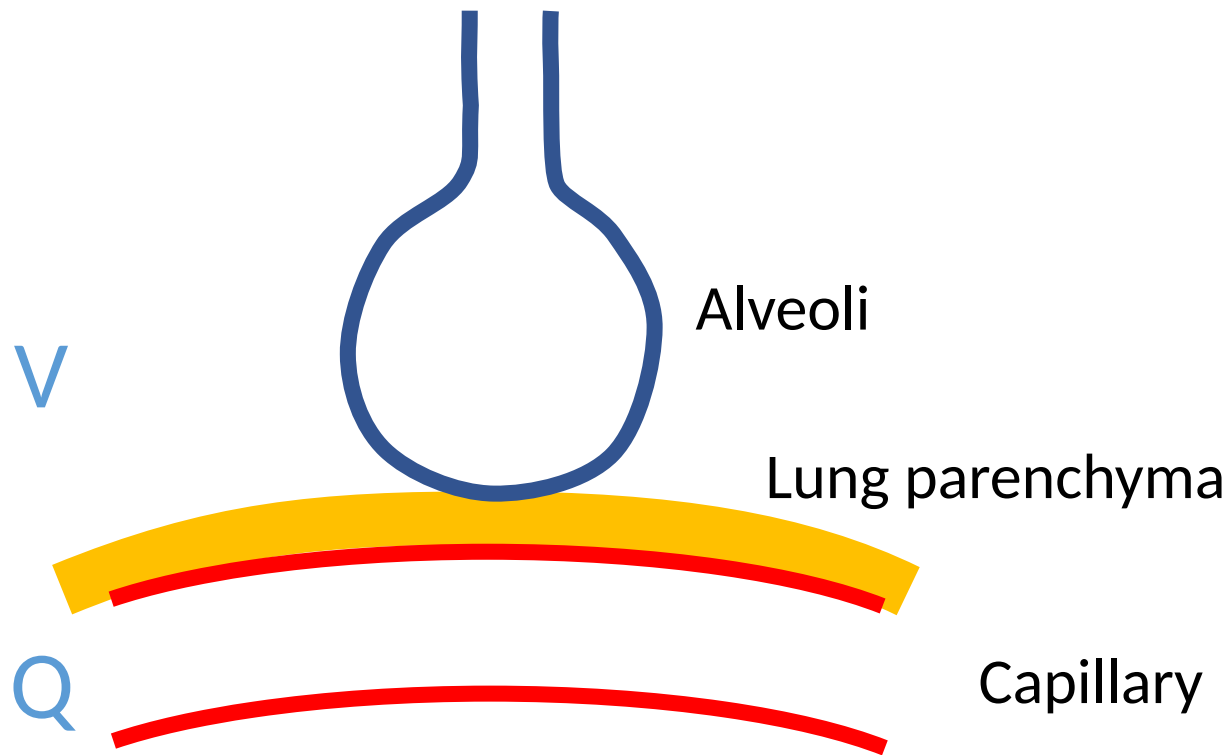
$$P_AO_2 = 110$$

$$P_AO_2 - P_aO_2 = 110 - 50 = 60$$

Elevated A-a gradient!

$$\begin{aligned}\text{Expected A-a gradient} &= \text{Age}/4 + 4 \\ &= 64/4 + 4 \\ &= 20\end{aligned}$$

Causes of an elevated A-a gradient



~~Shunt (anatomic, large alveolar filling process)~~

~~V/Q mismatch (obstructive lung disease, pulmonary vascular disease/PE)~~

~~↓ diffusion capacity~~

Non-anatomic shunts
are evident on CXR

Does not typically
happen acutely

CXR looks
like...



Image from Radiopaedia.org

GO BACK

The Age Old Question: ABG or VBG?

ABG: pH / $P_a\text{CO}_2$ / $P_a\text{O}_2$

VBG: pH / $P_v\text{CO}_2$

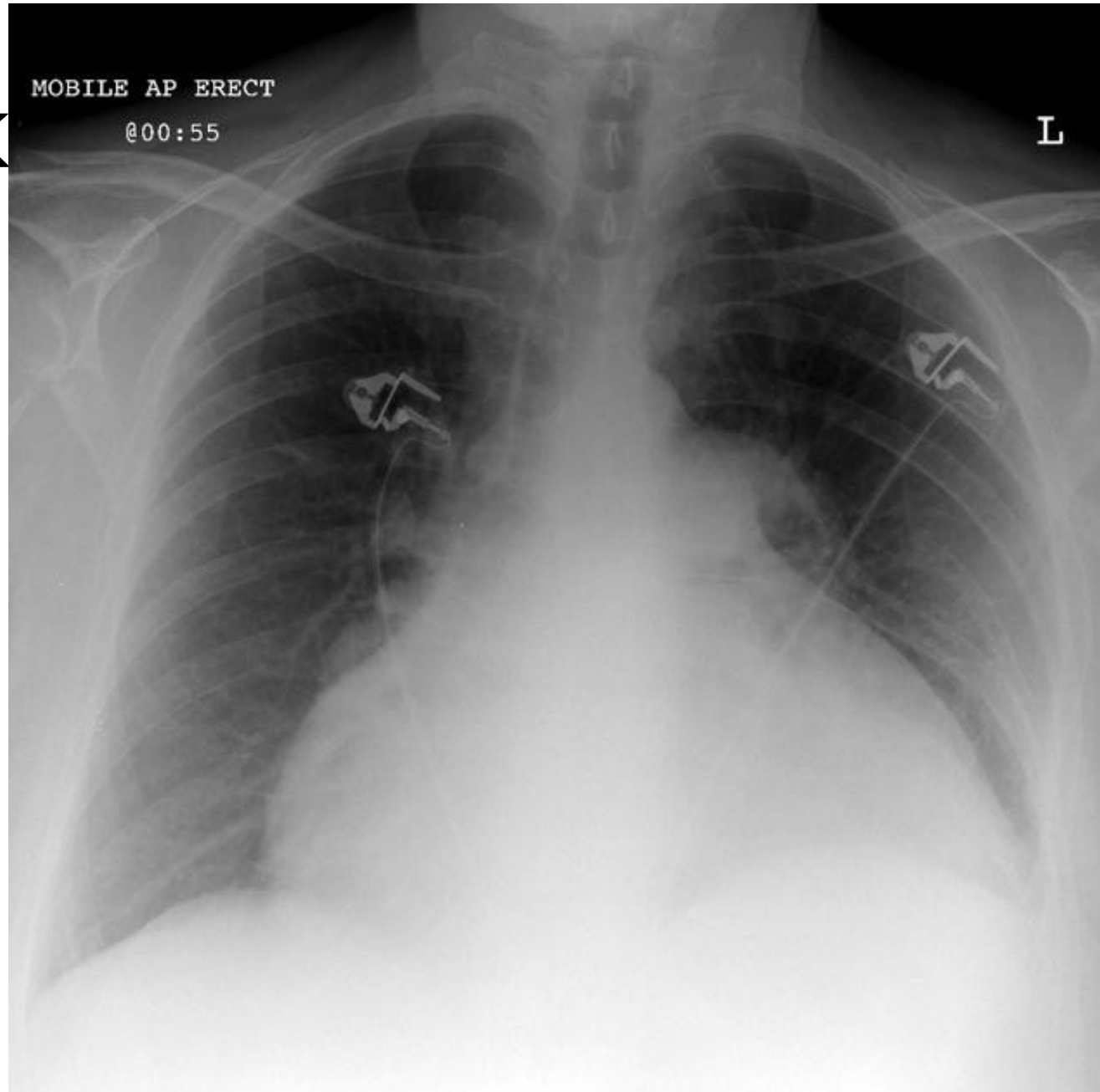


ABG pH = VBG pH + 0.04
ABG $P_a\text{CO}_2$ = VBG $P_v\text{CO}_2$ - 4

Get an ABG when you need the $P_a\text{O}_2$:

1. Calculate the A-a gradient
2. Confirming hypoxemia
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CXR looks like



GO BACK

ECG looks like...

GO BACK

